CASE STUDY QUESTION

SwiftMotors

Q1 Production Quality Report

Create a Production Quality Report grouped by Plant Location showing Total Units Produced, Total Defective Units, and Defect Rate.

- **Requirement:** Add a calculated column for Defect Rate = (Defective / Produced) * 100
- Apply Conditional Styling: Red for Defect Rate > 5%, Amber for 2–5%, Green for < 2%
- Add header with plant logo text, Footer with page number. Include prompts for Plant and Vehicle Type.

CASE STUDY 6

SwiftMotors & IBM Cognos Analytics

Automobile Manufacturing

Q1 Production Quality Report

Create a Production Quality Report grouped by Plant Location showing **Total Units Produced, Total Defective Units, and Defect Rate**

Requirement: Add a calculated column for Defect Rate = (Defective / Produced) * 100

- Apply Conditional Styling: Red for Defect Rate > 5%, Amber for 2–5%, Green for < 2%.
- Add header with plant logo text, Footer with page number. Include prompts for Plant and Vehicle Type.

Steps to Create the Report:

01 Set Up Report Layout

Design the report in IBM Cognos Analytics. Add a header with the SwiftMotors logo and title **"Production Quality Report"**.

02 Group Data and Add Columns

Group by Plant Location. Add columns for Total Units Produced, Total Defective Units.

03 Calculate Defect Rate

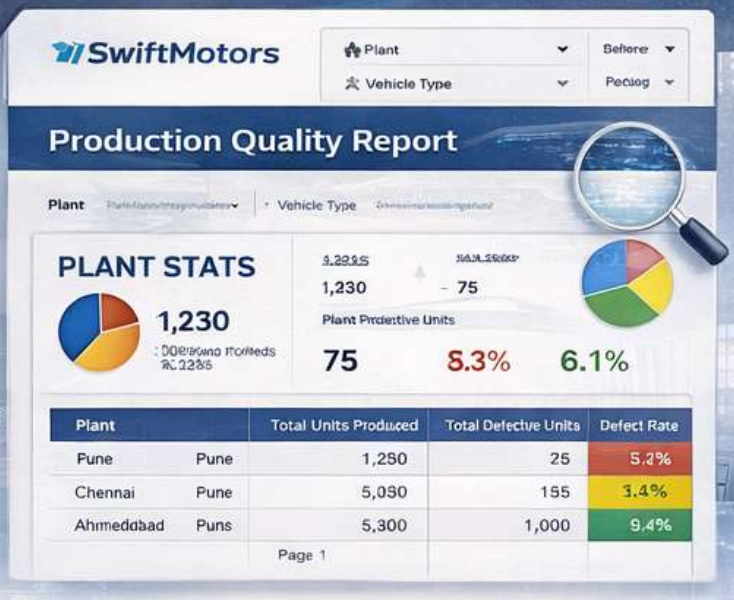
Create a calculated column for Defect Rate using the formula (Defective / Produced) * * 100.

Calculanprer	
Solarcd	2.3%
Cbarity	13%
6,30d	2.3%

04 Apply Conditional Styling

Style the Defect Rate column: **Red for Defect Rate > 5%, Amber for 2–5%, Green for < 2%.**

Plant	Total Units	Total Defective Units	Defect Rate
Pune	1,330	7,350	5.8%
Chennai	1,050	5,356	6.9%
Ahmedabad	1,050	5,366	5.4%



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IBM Cognos Analytics | * My CS 1

Edit | Report > Pages > Page1 | Page design | Properties

Insertable objects

Find

PINNED

Text item | Block | Table | List | Crosstab | Visualization

TEXTUAL

Text item | Query calculation | Layout calculation | Rich text item | Hyperlink | Date | Time | Page number | Row number | As-of-time expression

LAYOUT

DATA CONTAINER

PROMPTING

AUTHORED CONTROLS

ADVANCED

DEVELOPER

Swift Motors – Production Quality Report

Plant Location	Units Produced	(Units Defective / Units Produced)	Defect Rate	Units Defective	% of total (Units Defective)
<Plant Location>	<Units Produced>	<(Units Defective / Units Produced)>	<Defect Rate>	<Units Defective>	<% of total (Units Defective)>
	<Units Produced>	<(Units Defective / Units Produced)>	<Defect Rate>		<% of total (Units Defective)>
<Plant Location>	<Units Produced>	<(Units Defective / Units Produced)>	<Defect Rate>		<% of total (Units Defective)>
	<Units Produced>	<(Units Defective / Units Produced)>	<Defect Rate>		<% of total (Units Defective)>
<Units Defective>		**<(Units Defective / Units Produced)1>**			
<Units Defective> - Total			**<Total(Defect Rate)>**		
<Plant Location>	<Units Produced>	<(Units Defective / Units Produced)>	<Defect Rate>	<Units Defective>	<% of total (Units Defective)>
	<Units Produced>	<(Units Defective / Units Produced)>	<Defect Rate>		<% of total (Units Defective)>
<Plant Location>	<Units Produced>	<(Units Defective / Units Produced)>	<Defect Rate>		<% of total (Units Defective)>
	<Units Produced>	<(Units Defective / Units Produced)>	<Defect Rate>		<% of total (Units Defective)>
<Units Defective>		**<(Units Defective / Units Produced)1>**			
<Units Defective> - Total			**<Total(Defect Rate)>**		
Overall - Total	**<Total(Units Produced)>**	**<(Units Defective / Units Produced)1>**	**<Total(Defect Rate)>**	**<Total(Units Defective)>**	

1

Page footer

insert 0/0

CONDITIONAL
Conditional styles
Style variable

BOX
Border
Padding
Box type

COLOR & BACKGROUND
Background image
Background effects
Background color
Foreground color

FONT & TEXT
Font
Horizontal alignment
Vertical alignment
White space
Spacing & breaking
Direction & justification

POSITIONING
Size & overflow

MISCELLANEOUS
Classes

ap2.ca.analytics.ibm.com/bi/?perspective=authoring&id=i18A835F4537342FE88D300B011B432B0&objRef=i18A835F4537342FE88D300B011B432B0...

IBM Cognos Analytics | * My CS 1

Edit | Report > Prompt pages > **Prompt page1** | Page design | Properties

Insertable objects

Find

PINNED

Text item, Block, Table, List, Crosstab, Visualization

TEXTUAL

Text item, Query calculation, Layout calculation, Rich text item, Hyperlink, Date, Time, Page number, Row number, As-of-time expression

LAYOUT

DATA CONTAINER

PROMPTING

AUTHORED CONTROLS

ADVANCED

DEVELOPER

Swift Motors Plant Location

plant location

vehicle type

Cancel | < Back | Next > | Finish

Page body

insert 0/0

CONDITIONAL

Conditional styles

Style variable

BOX

Border

Padding

Box type

COLOR & BACKGROUND

Background image

Background effects

Background color

Foreground color

FONT & TEXT

Font

Horizontal alignment

Vertical alignment

White space

Spacing & breaking

Direction & justification

POSITIONING

Size & overflow

MISCELLANEOUS

Classes: Prompt page body

ap2.ca.analytics.ibm.com/bi/?perspective=authoring&closeWindowOnLastView=true

Maintenance: The upgrade is now complete. Click on More Info to see what actions may be necessary and to subscribe to future events

Dismiss More info

IBM Cognos Analytics My CS 1

HTML

Mode: Interactive View Filters

Swift Motors Plant Location

plant location

- [ ] Ahmedabad
- [x] Chennai
- [ ] Pune

Select all Deselect all

vehicle type

- ( ) EV
- (•) SUV
- ( ) Sedan

Deselect all

Cancel < Back Next > Finish

Swift Motors – Production Quality Report

Plant Location	Units Produced	(Units Defective / Units Produced)	Defect Rate	Units Defective	% of total (Units Defective)
Pune	1,439	0.03961084	3.96108409	57	30.65%
		0.03961084	3.96108409		30.65%
		0.03961084	3.96108409		30.65%
		0.03961084	3.96108409		30.65%
		0.03961084	3.96108409		30.65%
		0.03961084	3.96108409		30.65%
		0.03961084	3.96108409		30.65%
57		**0.08069164**			
57 - Total			**8.47448324**		
Ahmedabad	1,418	0.04513399	4.51339915	64	34.41%
		0.04513399	4.51339915		34.41%
		0.04513399	4.51339915		34.41%
		0.04513399	4.51339915		34.41%
		0.04513399	4.51339915		34.41%
		0.04513399	4.51339915		34.41%
		0.04513399	4.51339915		34.41%
64		**0.08069164**			
64 - Total			**8.47448324**		
Chennai	2,148	0.03026071	3.02607076	65	34.95%
		0.03026071	3.02607076		34.95%
		0.03026071	3.02607076		34.95%
		0.03026071	3.02607076		34.95%
		0.03026071	3.02607076		34.95%
		0.03026071	3.02607076		34.95%
		0.03026071	3.02607076		34.95%
65		**0.06593407**			
65 - Total			**3.02607076**		
Overall - Total	**5,005**	**0.03716284**	**11.500554**	**186**	

Swift Motors – Production Quality Report

Plant Location	Units Produced	(Units Defective / Units Produced)	Defect Rate	Units Defective	% of total (Units Defective)
Ahmedabad	1,714	0.04550758	4.55075846	78	32.50%
		0.04550758	4.55075846		32.50%
		0.04550758	4.55075846		32.50%
		0.04550758	4.55075846		32.50%
		0.04550758	4.55075846		32.50%
		0.04550758	4.55075846		32.50%
		0.04550758	4.55075846		32.50%
		0.04550758	4.55075846		32.50%
78		**0.0921659**			
78 - Total			**8.93672337**		
Pune	1,824	0.04385965	4.38596491	80	33.33%
		0.04385965	4.38596491		33.33%
		0.04385965	4.38596491		33.33%
		0.04385965	4.38596491		33.33%
		0.04385965	4.38596491		33.33%
		0.04385965	4.38596491		33.33%
		0.04385965	4.38596491		33.33%
		0.04385965	4.38596491		33.33%
		0.04385965	4.38596491		33.33%
80		**0.0921659**			
80 - Total			**8.93672337**		
Chennai	1,758	0.04664391	4.66439135	82	34.17%
		0.04664391	4.66439135		34.17%
		0.04664391	4.66439135		34.17%
		0.04664391	4.66439135		34.17%
		0.04664391	4.66439135		34.17%
		0.04664391	4.66439135		34.17%
		0.04664391	4.66439135		34.17%
82		**0.07777778**			
82 - Total			**9.05035627**		
Overall -	**5,296**	**0.04531722**	**13.60111473**	**240**	

Swift Motors – Production Quality Report

Plant Location	Units Produced	(Units Defective / Units Produced)	Defect Rate	Units Defective	% of total (Units Defective)
Total					

Swift Motors – Production Quality Report

Plant Location	Units Produced	(Units Defective / Units Produced)	Defect Rate	Units Defective	% of total (Units Defective)
Chennai	284	0.04929577	4.92957746	14	15.22%
		0.04929577	4.92957746		15.22%
		0.04929577	4.92957746		15.22%
		0.04929577	4.92957746		15.22%
14		**0.07042254**			
14 - Total			**16.46585976**		
Ahmedabad	689	0.05079826	5.07982583	35	38.04%
		0.05079826	5.07982583		38.04%
		0.05079826	5.07982583		38.04%
		0.05079826	5.07982583		38.04%
		0.05079826	5.07982583		38.04%
		0.05079826	5.07982583		38.04%
		0.05079826	5.07982583		38.04%
35		**0.09090909**			
35 - Total			**11.53628229**		
Pune	666	0.06456456	6.45645646	43	46.74%
		0.06456456	6.45645646		46.74%
		0.06456456	6.45645646		46.74%
		0.06456456	6.45645646		46.74%
		0.06456456	6.45645646		46.74%
		0.06456456	6.45645646		46.74%
		0.06456456	6.45645646		46.74%
43		**0.09230769**			
43 - Total			**6.45645646**		
Overall - Total	**1,639**	**0.05613179**	**16.46585976**	**92**	